

Kolmogorov-Smirnov-test.R

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empirical_CDF_and_Kolmogorov_Test

Instruction: Study the empirical cumulative distribution function (ECDF) and the Kolmogorov–Smirnov test. Test whether a dataset follows a given distribution, or whether two samples come from the same one. Visualize ECDFs (implemented by yourself) and interpret.

for following the project on your own clone by:

```
git clone https://github.com/Aschir/empirical_CDF_and_Kolmogorov_Test/
```

theory about Kolmogorov-Smirnov test (KS-Test)

One-sample Kolmogorov-Smirnov Test

The one-sample Kolmogorov–Smirnov (KS) test quantifies the difference between the empirical distribution of a dataset and a theoretical distribution and can be applied when your analysed data is iid.

Example:

Testing if $X \sim N(\mu, \sigma^2)$

A basketball player Mike, shoots each training 1000 times. he has 200 trainings and scores with a probability of $p=0.5$, does his score follow a normal distribution?

$$training \sim Binom(n = 1000, p = 0.5)$$

with 100 performed trainings:

```
set.seed(42)
# data
n = 1000
prob = 0.5
n_training = 100

X = rbinom(n_training, size = n, prob = prob)
```

$$H_0 : X \sim N(\mu = 500, \sigma^2)$$

$$H_1 : X \sim N(\mu = 500, \sigma^2)$$

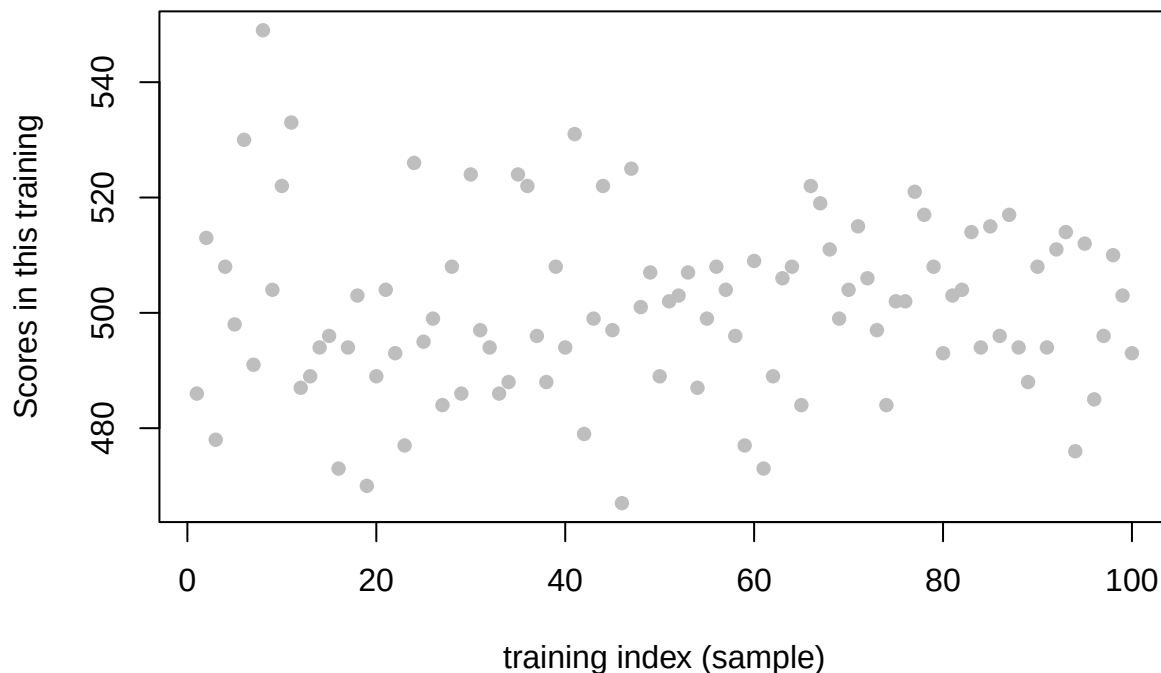
Let's have a look in it's distribution:

```
set.seed(42)

# for visualisation: 10% quantile and closest point to 10% quantile
q10 <- quantile(X, 0.1)
closest_q10 <- X[which.min(abs(X - q10))]

# ---- Plot 1: the distribution of X ----
plot(1:length(X), X,
     xlab = "training index (sample)",
     ylab = "Scores in this training",
     main = "Distribution of scores in the progress of trainings.",
     pch = 16,
     col="grey")
```

Distribution of scores in the progress of trainings.



```
# ---- Plot 2: Histogram + part shaded + theoretical curve ----
h = hist(X,
        breaks = 50,
```

```

    probability = TRUE,
    main = paste0("Distribution of scored points"),
    xlab = "Scores in a training",
    col = "lightgray",
    border = "white")

# --- Shade bars up to closest 10% quantile ---
for (i in 1:(length(h$breaks)-1)) {
  left  <- h$breaks[i]
  right <- h$breaks[i+1]
  mid   <- (left + right) / 2

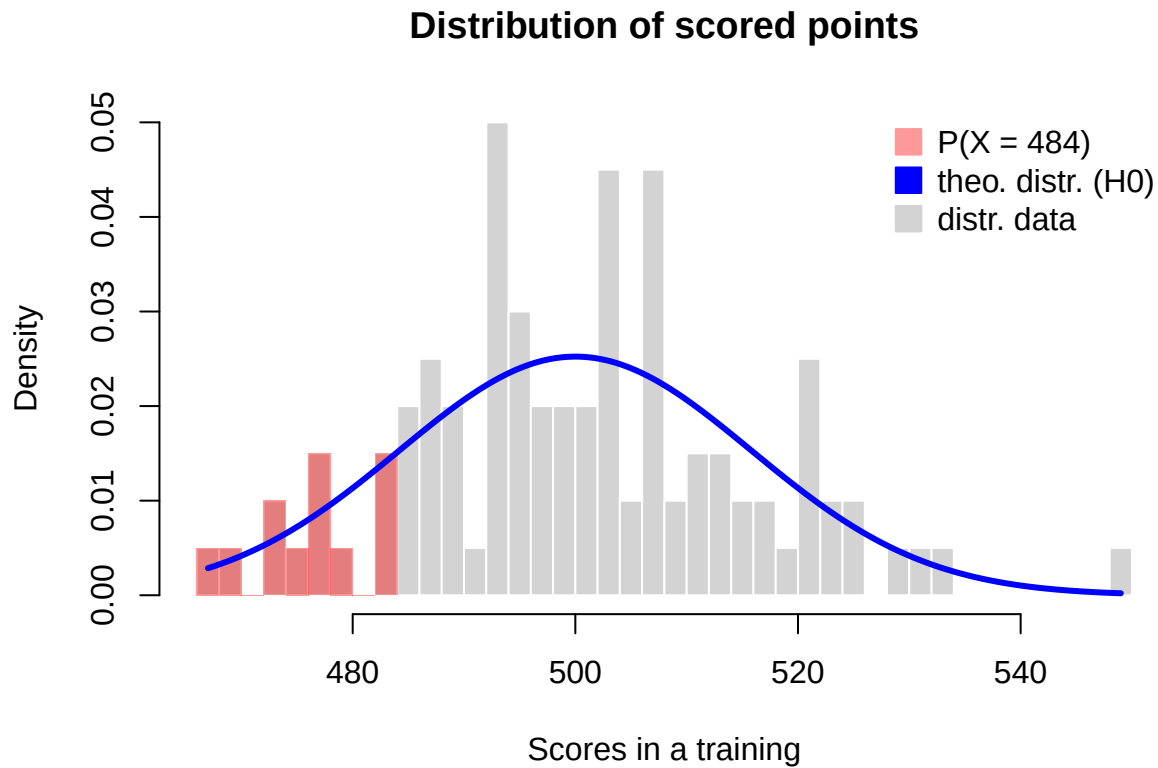
  if (mid <= closest_q10) {
    rect(left, 0, right, h$density[i],
         col = rgb(1, 0, 0, 0.4),
         border = NA)
  }
}

# --- Theoretical distribution ---
x_vals <- seq(min(X), max(X), length.out = 200)
theo_density <- dnorm(x_vals, mean = 500, sd = sqrt(250))

lines(x_vals, theo_density, col = "blue", lwd = 3)

# --- Legend ---
legend(
  "topright",
  legend = c(paste0("P(X  ", closest_q10, ")"),
             "theo. distr. (H0)",
             "distr. data"),
  pch     = 22,
  pt.cex  = 2,
  pt.bg   = c(rgb(1,0,0,0.4), "blue", "lightgray"), # fill colors
  col     = c(rgb(1,0,0,0.4), "blue", "lightgray"), # border colors
  bty     = "n"
)

```



In a general setting,

Null hypothesis: $H_0 : F_X = F_0$

Alternative hypothesis: $H_1 : F_X \neq F_0$

and our test statistic is:

$$D_n = \sup_x |F_n(x) - F(x)|$$

We make no assumptions on the parameter of the given distribution! Only the CDF

We Compute the CDF:

```
set.seed(42)

#CDF (ECDF) of the given DATA.
ecdf_X = ecdf(X) # we use the given func for the data

plot(ecdf_X,
     main = "Empirical CDF of X",
     xlab = "x",
     ylab = "F(x)",
     verticals = TRUE,
     do.points = FALSE)

# theoretical cdf in manual computation of the cdf
x_vals = seq(min(X), max(X), length.out = 200)
```

```

cdf_theo <- pnorm(x_vals, mean = 500, sd = sqrt(250))

lines(x_vals, cdf_theo, col = "blue", lwd = 2)

# Add the point seen in the previous data
y_q10 <- ecdf_X(closest_q10)

points(closest_q10, y_q10, col = rgb(1, 0, 0, 0.4), pch = 16)

# ---- Add vertical line for sup difference (Kolmogorov D) ----

# max difference computation
emp_vals <- ecdf_X(x_vals)
diffs <- abs(emp_vals - cdf_theo)
D <- max(diffs)

x_value_f <- x_vals[which.max(diffs)]
y_emp_at_f <- emp_vals[which.max(diffs)]
y_theo_at_f <- cdf_theo[which.max(diffs)]
eps <- 1e-8

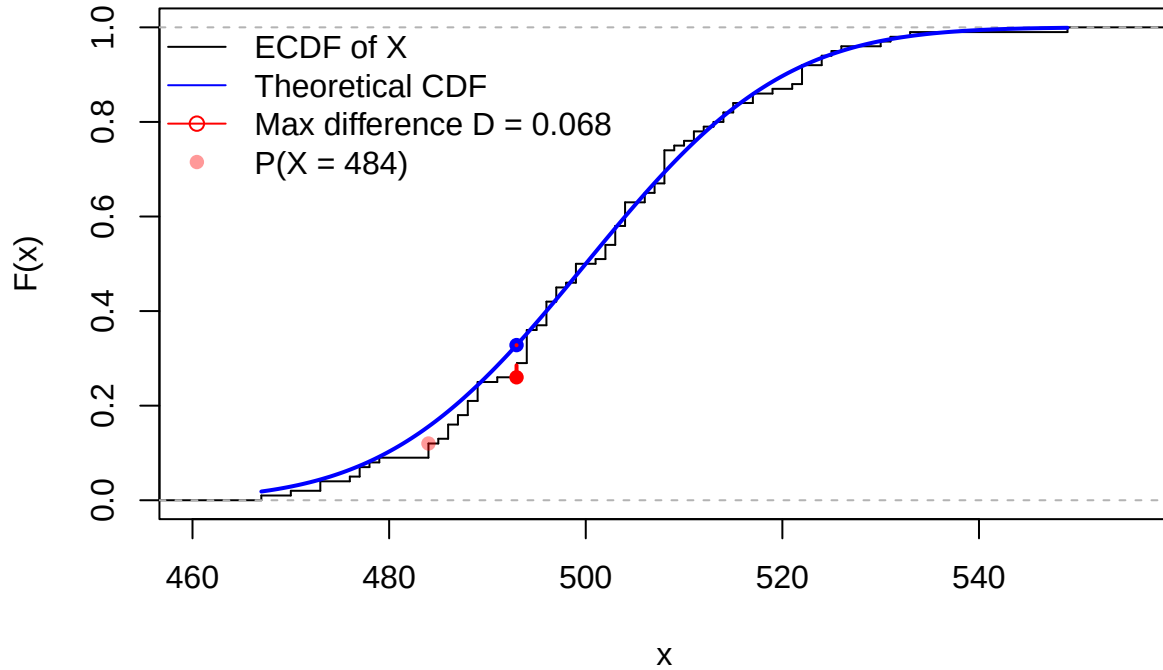
# draw the vertical line of max difference
#segments(x0 = x_value_f, y0 = y_emp_at_f,
#         x1 = x_value_f, y1 = y_theo_at_f,
#         col = "red", lwd = 2, lty = 2)

# mark max difference points in their distributions
points(x_value_f, y_emp_at_f, col="red", pch=16)
points(x_value_f, y_theo_at_f, col="blue", pch=16)
lines(x = c(x_value_f-eps, x_value_f+eps), y = c(y_emp_at_f, y_theo_at_f),
      col = "red", lwd = 2, lty = 2)

legend("topleft",
      legend = c("ECDF of X",
                  "Theoretical CDF",
                  paste0("Max difference D = ", round(D, 3)),
                  paste0("P(X ", closest_q10, ")")),
      col = c("black", "blue", "red", rgb(1,0,0,0.4)),
      lwd = c(1, 1, 1, NA),
      lty = c(1, 1, 1, NA),
      pch = c(NA, NA, 1, 16),
      bty = "n")

```

Empirical CDF of X



we used the maximum as a numerical approximation of the supremum.

the supremum would consider all points not only the ones in the dataset.

the maximum deviation difference is not normal distributed but follows a Kolmogorov-Smirnov deviations-distribution defined by:

The maximum deviation D_n is **not normally distributed**. Instead, it follows the **Kolmogorov distribution**, which has the cumulative distribution function:

$$P(\sqrt{n}D_n \leq \lambda) = 1 - 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 \lambda^2}$$

which is in r:

```
ks_pvalue <- function(D, n, maxK=1000){
  lambda <- sqrt(n) * D
  k <- 1:maxK
  pval <- 2 * sum((-1)^(k-1) * exp(-2 * k^2 * lambda^2))
  pval <- min(max(pval, 0), 1)
  return(pval)
}
```

now we compute the p-value: and form our decision boundary:

$$P(D_n \geq D_{\text{obs}} \mid H_0)$$

```
p_value <- ks_pvalue(D, length(X))
p_value
```

```
## [1] 0.7431522
```

```
# Decision based on alpha
alpha <- 0.05
if(p_value < alpha){
  cat("Reject H0: the data does not follow the theoretical distribution.\n")
} else {
  cat("Fail to reject H0: no evidence against the theoretical distribution.\n")
}
```

```
## Fail to reject H0: no evidence against the theoretical distribution.
```

we can compare our computation with the builtin-method:

the results are not very accurate since we used the maximum instead of the minimum and for sakes of explainability an discrete distribution function instead of an continious function

```
# Built-in KS test
ks_result <- ks.test(X, "pnorm", mean = 500, sd = sqrt(250))
```

```
## Warning in ks.test.default(X, "pnorm", mean = 500, sd = sqrt(250)): ties should
## not be present for the one-sample Kolmogorov-Smirnov test
```

```
ks_result
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: X
## D = 0.068985, p-value = 0.7281
## alternative hypothesis: two-sided
```